LESSON PLAN

AIR NAVIGATION (21)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	15
Total No. of Lesson Plans	18	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Practical on identifying various components and analyzing of Wiring diagram manual.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Practical on identifying various components and analyzing of Wiring diagram manual.
5. Specific Objectives	21.1.10.1 Trainee will be able to explain how to carry out DI / DR on aircraft system.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Identification of components on wiring diagram; verification of circuit paths; safety precautions regarding electrical isolation; instructor supervision during diagram analysis.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary purpose of a wiring diagram manual?
(b) List three common symbols found in aircraft electrical schematics.
(c) Why is it critical to verify power isolation before analyzing a circuit?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Identification of standard electrical symbols • Interpretation of wire color codes • Tracing circuit paths from power source • Locating component reference designations • Understanding circuit breaker and fuse ratings	L & Ds	PPT, IB	15 min
(b)	• Analysis of component interconnections • Verification of ground and power points • Identification of switch positions and functions • Review of circuit protection devices • Cross-referencing with system logic	L & Ds	PPT, PDF	20 min
(c)	• Execution of DI/DR procedures • Documentation of findings • Troubleshooting common diagram errors • Safety checks during analysis • Final review of circuit integrity	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the step-by-step process for performing a DI/DR on an aircraft system.
(b) How do you verify that a circuit is correctly grounded according to the diagram?
(c) What safety precautions must be taken before analyzing a live or potentially live circuit?

12. Home Assignment and Follow Up	Prepare a brief report identifying five key components in the next system's wiring diagram and explain their function, bridging the knowledge required for LP 02.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Identification / removal and installation of various system components.	Refer to the master syllabus reference list	LP 16

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min